

Quantum Kerr OAM

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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2), \quad [\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)] = [\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)] = 0 \quad (1)$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) d\mathbf{r}. \quad (2)$$

We then have

$$[a(v_1), a(v_2)^\dagger] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r}) v_2(\mathbf{r}) d\mathbf{r} = \langle v_1 | v_2 \rangle, \quad (3)$$

while $[a(v_1), a(v_2)] = [a(v_1)^\dagger, a(v_2)^\dagger] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}^2} v^*(\mathbf{r}) \mathcal{E}(\mathbf{r}) + v(\mathbf{r}) \mathcal{E}^\dagger(\mathbf{r}) d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \quad (4)$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i \operatorname{Im} \langle v_1 | v_2 \rangle, \quad (5)$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2 \Delta X(v_2)^2 \geq \frac{|\operatorname{Im} \langle v_1 | v_2 \rangle|^2}{4}. \quad (6)$$

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One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i \operatorname{Re} \langle v_1 | v_2 \rangle \quad (7)$$

and

$$\Delta X(v_1)^2 \Delta P(v_2)^2 \geq \frac{|\operatorname{Re} \langle v_1 | v_2 \rangle|^2}{4}. \quad (8)$$

We are interested in observables of the form

$$C(v_1, v_2) = \frac{1}{2} \langle \{X(v_1), X(v_2)\} \rangle - \langle X(v_1) \rangle \langle X(v_2) \rangle = \operatorname{Re} \left[\frac{\langle v_1 | v_2 \rangle}{2} + f(v_1, v_2) + g(v_1, v_2) \right], \quad (9)$$

where we have defined

$$\begin{aligned} f(v_1, v_2) &= \langle a(v_1) a(v_2) \rangle - \langle a(v_1) \rangle \langle a(v_2) \rangle \\ g(v_1, v_2) &= \langle a(v_2)^\dagger a(v_1) \rangle - \langle a(v_2)^\dagger \rangle \langle a(v_1) \rangle \end{aligned} \quad (10)$$

This formulation is useful because f and g are normally ordered expectation values, which can be directly obtained from stochastic simulations in the positive P representation. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1, v_2 + \lambda v_3) &= f(v_1, v_2) + \lambda^* f(v_1, v_3) \\ f(v_1, v_2) &= f(v_2, v_1) \\ g(v_1, v_2 + \lambda v_3) &= g(v_1, v_2) + \lambda g(v_1, v_3) \\ g(v_1, v_2) &= g(v_2, v_1)^* \end{aligned} \quad (11)$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\sigma_{jk} = \frac{1}{2} \langle \{Z_j, Z_k\} \rangle - \langle Z_j \rangle \langle Z_k \rangle$ where $\mathbf{Z} = (X(v), P(v))$. We observe that $g(v, v) \in \mathbb{R}$, so we get

$$\sigma = \begin{pmatrix} \frac{1}{2} + g(v, v) & 0 \\ 0 & \frac{1}{2} + g(v, v) \end{pmatrix} + \begin{pmatrix} \operatorname{Re} f(v, v) & -\operatorname{Im} f(v, v) \\ -\operatorname{Im} f(v, v) & -\operatorname{Re} f(v, v) \end{pmatrix}. \quad (12)$$

Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi}f(v, v)$. Therefore, we can diagonalize σ by choosing $\phi = [\arg f(v, v) \pm \pi]/2$. We then get eigenvalues

$$\lambda_{\pm} = \frac{1}{2} + g(v, v) \pm |f(v, v)|. \quad (13)$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_- , while the one rotated along the axis $\phi + \pi/2$ has maximum variance λ_+ . Furthermore, we have squeezing if $|f(v, v)| > g(v, v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2 , i.e., $\langle v_j | v_k \rangle = \delta_{jk}$. The Duan criterion states that if

$$D = \left\langle \Delta [X(v_1) + X(v_2)]^2 \right\rangle + \left\langle \Delta [P(v_1) - P(v_2)]^2 \right\rangle < 2, \quad (14)$$

then the states are entangled.

Let us define $v_{\pm} = v_1 \pm v_2$. We then have

$$D = \langle \Delta X(v_+)^2 \rangle + \langle \Delta P(v_-)^2 \rangle. \quad (15)$$

Although $v_{\pm}$ is not normalized, (9) still holds. We then obtain

$$\begin{aligned} D &= 2 + g(v_+, v_+) + g(v_-, v_-) + \text{Re} [f(v_+, v_+) - f(v_-, v_-)] \\ &= 2 [1 + g(v_1, v_1) + g(v_2, v_2)] + 4 \text{Re} f(v_1, v_2) \end{aligned} \quad (16)$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_2 by an angle $\phi = [\arg f(v_1, v_2) \pm \pi] / 2$. We get an optimal possible violation of

$$D_{\text{opt}} = 2 [1 + g(v_1, v_1) + g(v_2, v_2)] - 4 |f(v_1, v_2)| \quad (17)$$

II. THE KERR SYSTEM

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z \mathcal{E}(\mathbf{r}, z) + \nabla^2 \mathcal{E}(\mathbf{r}, z) = -g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (18)$$

Let us decompose $\mathcal{E} = \mathcal{E}_0 + \delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z \mathcal{E}_0(\mathbf{r}, z) + \nabla^2 \mathcal{E}_0(\mathbf{r}, z) = 0, \quad \mathcal{E}_0(\mathbf{r}, 0) = \mathcal{E}(\mathbf{r}, 0). \quad (19)$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r}, z) + \nabla^2 u(\mathbf{r}, z) = 0$. Then, the annihilation operator corresponding to mode u is given by

$$a_u(z) = \int u^*(\mathbf{r}, z) \mathcal{E}(\mathbf{r}, z) d\mathbf{r} = a_{u,0} + \delta a_u(z) \quad (20)$$

where

$$a_{u,0} = \int u^*(\mathbf{r}, z) \mathcal{E}_0(\mathbf{r}, z) d\mathbf{r} \quad (21)$$

is independent of z (take a z derivative, use the equations of motion and integrate by parts) and

$$\delta a_u(z) = \int u^*(\mathbf{r}, z) \delta \mathcal{E}(\mathbf{r}, z) d\mathbf{r} \quad (22)$$

Our initial state will be a coherent one:

$$|\alpha, u\rangle = D(\alpha, u) |0\rangle = \exp \left(\alpha a_{u,0}^\dagger - \alpha^* a_{u,0} \right) |0\rangle. \quad (23)$$

This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r}, z) |\alpha, u\rangle = \alpha u(\mathbf{r}, z) |\alpha, u\rangle \quad (24)$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z \delta\mathcal{E}(\mathbf{r}, z) + \nabla^2 \delta\mathcal{E}(\mathbf{r}, z) = -g\mathcal{E}^\dagger(\mathbf{r}, z)\mathcal{E}^2(\mathbf{r}, z). \quad (25)$$

Evaluating this at $z = 0$ gives us

$$2ik\partial_z \delta\mathcal{E}(\mathbf{r}, 0) = -g\mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \quad (26)$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r}, z) \approx \frac{igz}{2k} \mathcal{E}_0^\dagger(\mathbf{r}, 0)\mathcal{E}_0^2(\mathbf{r}, 0) \approx \frac{igz}{2k} \mathcal{E}_0^\dagger(\mathbf{r}, z)\mathcal{E}_0^2(\mathbf{r}, z) \quad (27)$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r}, z) \approx \mathcal{E}_0(\mathbf{r}, z) + \frac{igz}{2k} \mathcal{E}_0^\dagger(\mathbf{r}, z)\mathcal{E}_0^2(\mathbf{r}, z) \quad (28)$$

This expression will be the basis of our further developments.

A. Expectation value of field

Taking expectation values, we arrive at

$$\langle \mathcal{E}(\mathbf{r}, z) \rangle \approx \alpha u(\mathbf{r}, z) + \frac{igz|\alpha|^2\alpha}{2k} |u(\mathbf{r}, z)|^2 u(\mathbf{r}, z). \quad (29)$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v \rangle = \alpha \langle v|u \rangle + \frac{igz|\alpha|^2\alpha}{2k} \langle v, u|u, u \rangle \quad (30)$$

In the above expression we have introduced the first spatial overlap

$$\langle v|u \rangle = \int v^*(\mathbf{r}, z) u(\mathbf{r}, z) d\mathbf{r} \quad (31)$$

and the second spatial overlap

$$\langle v_1, v_2|u_1, u_2 \rangle = \int v_1^*(\mathbf{r}, z) v_2^*(\mathbf{r}, z) u_1(\mathbf{r}, z) u_2(\mathbf{r}, z) d\mathbf{r}. \quad (32)$$

When $u = u_{0l}$ is a Laguerre-Gaussian mode with radial order $p = 0$ and topological charge l , we have that

$$|u_{0l}(\mathbf{r}, z)| u_{0l}^2(\mathbf{r}, z) = \sum_{q=0}^{|l|} C_{ql} \bar{u}_{ql}(\mathbf{r}, z) \quad (33)$$

where $\bar{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v \rangle = \alpha \langle v|u_{0,l} \rangle + \frac{igz|\alpha|^2\alpha}{2k} \sum_{q=0}^{|l|} C_{ql} \langle v|\bar{u}_{ql} \rangle \quad (34)$$

In particular, when $v = \bar{u}_{ql}$, we have that

$$\langle a_{\bar{u}_{ql}} \rangle = \begin{cases} \alpha \langle \bar{u}_{ql}|u_{0,l} \rangle + \frac{igz|\alpha|^2\alpha C_{ql}}{2k} & q \leq |l| \\ \alpha \langle \bar{u}_{ql}|u_{0,l} \rangle & q > |l| \end{cases}. \quad (35)$$

B. Expectation value of number

The product of the field at two points is

$$\mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \delta \mathcal{E}(\mathbf{r}_2, z) + \delta \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \quad (36)$$

where we have only kept terms linear in $\delta \mathcal{E}$.

Close to $z = 0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0^2(\mathbf{r}_2, z) - \mathcal{E}_0^\dagger(\mathbf{r}_1, z)^2 \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \right] \quad (37)$$

Evaluating this at our initial state gives us

$$\langle \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle \approx |\alpha|^2 u^*(\mathbf{r}_1, z) u(\mathbf{r}_2, z) \left\{ 1 + \frac{igz|\alpha|^2}{2k} \left[|u(\mathbf{r}_2, z)|^2 - |u(\mathbf{r}_1, z)|^2 \right] \right\} \quad (38)$$

The number operator of mode v can be written in terms of this:

$$n_v = a_v^\dagger a_v = \int v(\mathbf{r}_1) v^*(\mathbf{r}_2) \mathcal{E}^\dagger(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) d\mathbf{r}_1 d\mathbf{r}_2 \quad (39)$$

The expectation value is then

$$\langle n_v \rangle \approx |\alpha \langle v|u \rangle|^2 + \frac{gz|\alpha|^4}{k} \text{Im} \langle v|u \rangle \langle u, u|u, v \rangle \approx |\langle a_v \rangle|^2. \quad (40)$$

When both u and v are arbitrary Laguerre-Gaussian modes (any p, l, w), we have that $\text{Im} \langle v|u \rangle \langle u, u|u, v \rangle = 0$, so there is no first order correction to the occupation number.

C. Quadratures

For the Kerr system, we have

$$\begin{aligned}\langle a_{v_2}^\dagger a_{v_1} \rangle &= \int v_2(\mathbf{r}_2) v_1^*(\mathbf{r}_1) \langle \mathcal{E}^\dagger(\mathbf{r}_2, z) \mathcal{E}(\mathbf{r}_1, z) \rangle d\mathbf{r}_1 d\mathbf{r}_2 \\ &\approx |\alpha|^2 \langle v_1|u \rangle \langle u|v_2 \rangle + \frac{gz|\alpha|^4}{k} \text{Im} \langle v_1|u \rangle \langle u, u|u, v_2 \rangle \\ &\approx \langle a_{v_2}^\dagger \rangle \langle a_{v_1} \rangle\end{aligned}\quad (41)$$

So the last two terms in (9) do not contribute. On the other hand, we have that

$$\begin{aligned}\mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \mathcal{E}_0(\mathbf{r}_1, z) \delta \mathcal{E}(\mathbf{r}_2, z) + \delta \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \\ &\approx \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \frac{igz}{2k} \left[\mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) \right] \\ &= \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \frac{igz}{2k} \left[\mathcal{E}_0^\dagger(\mathbf{r}_2, z) \mathcal{E}_0(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_2, z) + \mathcal{E}_0^\dagger(\mathbf{r}_1, z) \mathcal{E}_0^2(\mathbf{r}_1, z) \mathcal{E}_0(\mathbf{r}_2, z) + \delta(\mathbf{r}_1 - \mathbf{r}_2) \mathcal{E}_0^2(\mathbf{r}_2, z) \right]\end{aligned}\quad (42)$$

The appearance of the delta function here is important, because now

$$\langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle \approx \alpha^2 u(\mathbf{r}_1, z) u(\mathbf{r}_2, z) \left[1 + \frac{igz|\alpha|^2}{2k} \left[|u(\mathbf{r}_2, z)|^2 + |u(\mathbf{r}_1, z)|^2 \right] \right] + \frac{igz\alpha^2}{2k} \delta(\mathbf{r}_1 - \mathbf{r}_2) u^2(\mathbf{r}_2, z) \quad (43)$$

$$\begin{aligned}\langle a_{v_1} a_{v_2} \rangle &= \int v_1^*(\mathbf{r}_1) v_2^*(\mathbf{r}_2) \langle \mathcal{E}(\mathbf{r}_1, z) \mathcal{E}(\mathbf{r}_2, z) \rangle d\mathbf{r}_1 d\mathbf{r}_2 \\ &\approx \alpha^2 \langle v_1|u \rangle \langle v_2|v_1 \rangle + \frac{igz|\alpha|^2\alpha^2}{2k} (\langle v_1|u \rangle \langle v_2, u|u, u \rangle + \langle v_2|u \rangle \langle v_1, u|u, u \rangle) + \frac{igz\alpha^2}{2k} \langle v_1, v_2|u, u \rangle \\ &\approx \langle a_{v_2} \rangle \langle a_{v_1} \rangle + \frac{igz\alpha^2}{2k} \langle v_1, v_2|u, u \rangle\end{aligned}\quad (44)$$

We then finally get that

$$C(v_1, v_2) \approx \frac{\langle v_1|v_2 \rangle}{2} - \frac{gz\alpha^2}{2k} \text{Im} \langle v_1, v_2|u, u \rangle, \quad (45)$$

where we have chosen to measure phases with respect to the input state ($\text{Im} \alpha = 0$).

Let us calculate the matrix $C_{jk} = \langle Z_j Z_k \rangle - \langle Z_j \rangle \langle Z_k \rangle$ with $Z = X_{v_1}, P_{v_1}, X_{v_2}, P_{v_2}$, in this order. We have that

$$C = \begin{pmatrix} C(v_1, v_1) & C(v_1, -iv_1) & C(v_1, v_2) & C(v_1, -iv_2) \\ C(-iv_1, v_1) & C(-iv_1, -iv_1) & C(-iv_1, v_2) & C(-iv_1, -iv_2) \\ C(v_2, v_1) & C(v_2, -iv_1) & C(v_2, v_2) & C(v_2, -iv_2) \\ C(-iv_2, v_1) & C(-iv_2, -iv_1) & C(-iv_2, v_2) & C(-iv_2, -iv_2) \end{pmatrix} \quad (46)$$

We now assume that $\langle v_1|v_2 \rangle = 0$. We then have $C = C_0 + \delta C$ where

$$C_0 = \frac{1}{2} \begin{pmatrix} 1 & -i & 0 & 0 \\ i & 1 & 0 & 0 \\ 0 & 0 & 1 & -i \\ 0 & 0 & i & 1 \end{pmatrix}; \quad \delta C = \frac{gz\alpha^2}{2k} \begin{pmatrix} -\text{Im} \langle v_1, v_1|u, u \rangle & -\text{Re} \langle v_1, v_1|u, u \rangle & -\text{Im} \langle v_1, v_2|u, u \rangle & -\text{Re} \langle v_1, v_2|u, u \rangle \\ -\text{Re} \langle v_1, v_1|u, u \rangle & \text{Im} \langle v_1, v_1|u, u \rangle & -\text{Re} \langle v_1, v_2|u, u \rangle & \text{Im} \langle v_1, v_2|u, u \rangle \\ -\text{Im} \langle v_1, v_2|u, u \rangle & -\text{Re} \langle v_1, v_2|u, u \rangle & -\text{Im} \langle v_2, v_2|u, u \rangle & -\text{Re} \langle v_2, v_2|u, u \rangle \\ -\text{Re} \langle v_1, v_2|u, u \rangle & \text{Im} \langle v_1, v_2|u, u \rangle & -\text{Re} \langle v_2, v_2|u, u \rangle & \text{Im} \langle v_2, v_2|u, u \rangle \end{pmatrix} \quad (47)$$

At this point, it becomes clear that the overlaps of the form $\langle v_1, v_2|u, u \rangle$ are the ones that dominate this sort of process. On the other hand, the generation of radial orders comes from the properties of the different overlap $\langle v, u|u, u \rangle$, which appears in (30). It is therefore hard to see any special implication of the selection rules here.

Let us restrict our attention to vortices: $u(\mathbf{r}) = u_r(r) e^{il\phi}$; $v_j(\mathbf{r}) = v_{j,r}(r) e^{il_j\phi + \pi/4}$ where the radial functions are real valued. We further assume that $l_1 + l_2 = 2l$. Then, all the real terms in δC disappear and we get

$$\delta C = \frac{gz\alpha^2}{2k} \begin{pmatrix} -R_{11}\delta_{l_1, l_2} & 0 & -R_{12} & 0 \\ 0 & R_{11}\delta_{l_1, l_2} & 0 & R_{12} \\ -R_{12} & 0 & -R_{22}\delta_{l_1, l_2} & 0 \\ 0 & R_{12} & 0 & R_{22}\delta_{l_1, l_2} \end{pmatrix}, \quad R_{jk} = 2\pi \int_0^\infty v_{r,j}(r) v_{r,k}(r) u_r^2(r) dr \quad (48)$$

When $l_1 \neq l_2$, the variance in each mode does not change, but we see correlations between the quadratures of different modes induced by R_{12} . When $l_1 = l_2$, we also see maximum squeezing along X_{v_j} and antisqueezing along P_{v_j} .

The EPR correlations are given by

$$\langle \Delta(X_{v_1} \pm X_{v_2})^2 \rangle + \langle \Delta(P_{v_1} \mp P_{v_2})^2 \rangle = 2 \mp \frac{gz\alpha^2}{k} R_{12} \quad (49)$$

Then, the state is entangled according to the Duan criterion [1].

III. NUMERICAL CONSIDERATIONS

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (18), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = \sqrt{\Delta A} \mathcal{E}(\mathbf{r}, z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}), \tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger] \approx \delta_{\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'}, \quad (50)$$

while the equation of motion reads

$$i\partial_{\tilde{z}} \tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = -\frac{1}{4} \tilde{\nabla}^2 \tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}} \tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}}, \tilde{z}) \tilde{\mathcal{E}}^2(\tilde{\mathbf{r}}, \tilde{z}) \quad (51)$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [2])

$$\begin{aligned} i\partial_{\tilde{z}} \psi_1(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4} \tilde{\nabla}^2 \psi_1(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}} \psi_2(\tilde{\mathbf{r}}, \tilde{z}) \psi_1^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}} \psi_1(\tilde{\mathbf{r}}, \tilde{z}) \xi_1(\tilde{\mathbf{r}}, \tilde{z}) \\ -i\partial_{\tilde{z}} \psi_2(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4} \tilde{\nabla}^2 \psi_2(\tilde{\mathbf{r}}, \tilde{z}) - \frac{g}{4\Delta\tilde{A}} \psi_1(\tilde{\mathbf{r}}, \tilde{z}) \psi_2^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}} \psi_2(\tilde{\mathbf{r}}, \tilde{z}) \xi_2(\tilde{\mathbf{r}}, \tilde{z}) \end{aligned} \quad (52)$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\langle \xi_j(\tilde{\mathbf{r}}, \tilde{z}) \xi_k(\tilde{\mathbf{r}}', \tilde{z}') \rangle = 0, \quad \langle \xi_j(\tilde{\mathbf{r}}, \tilde{z}) \xi_k(\tilde{\mathbf{r}}', \tilde{z}') \rangle = \delta_{jk} \delta_{\tilde{\mathbf{r}}, \tilde{\mathbf{r}}} \delta_{\tilde{z}, \tilde{z}'} \quad (53)$$

For a coherent initial state in the mode u , we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha u(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^* u^*(\tilde{\mathbf{r}}, 0)$.

The main quantities we are interested, f , and g , are then expressed as

$$f(v_1, v_2) = \int_{\mathbb{R}^2 \times \mathbb{R}^2} v_1^*(\mathbf{r}_1) v_2^*(\mathbf{r}_2) \langle \psi_1(\mathbf{r}_1) \psi_1(\mathbf{r}_2) \rangle d\mathbf{r}_1 d\mathbf{r}_2 \quad (54)$$

$$g(v_1, v_2) = \int_{\mathbb{R}^2 \times \mathbb{R}^2} v_1^*(\mathbf{r}_1) v_2(\mathbf{r}_2) \langle \psi_1(\mathbf{r}_1) \psi_2(\mathbf{r}_2) \rangle d\mathbf{r}_1 d\mathbf{r}_2 \quad (55)$$

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